4.11 Human Immunodeficiency Virus (HIV) Related Opportunistic Infections

PAEDIATRICS

CHAPTER 4.11	HUMAN IMMUNODEFICIENCY VIRUS-RELATED OPPORTUNISTIC INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Recommended	**Alternative**	
***Pneumocystis Jirovecii* (*Carinii*) Pneumonia (PJP/PCP)**			
Treatment	Co-trimoxazole 5 mg/kg (Trimethoprim component) IV/PO q6H for 21 days		• If PaO2 < 70 mmHg in room air, start either: • Prednisolone 1.5 - 2 mg/kg/day in 2 divided doses, then tail down by 1 mg/kg/day every 5 days over 10 - 12 days • Dexamethasone 0.3 - 0.5 mg/kg IM/IV q6H for 5 days • Methylprednisolone
Primary prophylaxis	Co-trimoxazole 4 mg/kg (Trimethoprim component) PO q24H. **Alternative regimen:** Co-trimoxazole 150 mg/m^2/day		• Primary prophylaxis starts from 6 weeks old in HIV-exposed infants until status is established.

Infections and Likely Organisms	Suggested Treatment		Notes
	Recommended	**Alternative**	
	(Trimethoprim component) in 2 divided doses, given on alternate days (Monday, Wednesday, Friday). <u>Under 6 months old:</u> Co-trimoxazole 20 mg (Trimethoprim component) PO q24H		
***Toxoplasma gondii* encephalitis**			
Treatment	Pyrimethamine 2 mg/kg/day (D1 & D2) PO loading dose, then 1 mg/kg/day for 2 - 6 months, then 1 mg/kg/day 3 times per week, to complete 12 months total duration +		• Transmission of toxoplasma to infants by mother with HIV is uncommon in early pregnancy (< 10%). However, the risk increases sharply if the mother has toxoplasma infection during the last few weeks of pregnancy. Toxoplasma encephalitis in HIV-infected children is uncommon. Congenital toxoplasmosis is diagnosed by presence in neonatal blood of toxoplasma specific IgA, IgM in the first 6 month of life; or persistence of toxoplasma specific IgG beyond 6-month-old. • Co-trimoxazole NOT to be used in children as treatment. • *Continue for 1 week after discontinuation of Pyrimethamine.

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	Sulfadiazine 50 mg/kg/dose PO q12H for 12 months + Folinic Acid* 10 - 25 mg PO q24H with each dose of Pyrimethamine		
***Cryptococcal* meningitis**			
Induction phase	Amphotericin B Deoxycholate 0.7-1.5 mg/kg IV q24H over 4-6 hours for at least 2 weeks + Flucytosine 25 mg/kg PO q6H for at least 2 weeks		
Consolidation phase	Fluconazole 6 mg/kg IV/PO q12H (800 mg/day) for 8 - 10 weeks **OR**		

Infections and Likely Organisms	Suggested Treatment		Notes
	Recommended	**Alternative**	
	Itraconazole 5 mg/kg (max 200 mg) q12H PO for 8 weeks		
Secondary prophylaxis	Fluconazole 3 - 6 mg/kg PO q24H (max 200 mg/day)		• May be stopped when CD4 > 200.
CMV retinitis			
Treatment	Ganciclovir 5 mg/kg IV q12H for 2 - 3 weeks **OR** Valganciclovir 16 mg/kg PO q12H for 2 - 3 weeks	Foscarnet* 60 mg/kg IV q8H for 14 - 21 days then, 90 - 120 mg/kg IV q24H for chronic suppressive therapy	• *Foscarnet is not available in UMMC formulary.
***Mycobacterium avium* complex (MAC)**			
Treatment	Ethambutol 15 - 25 mg/kg PO q24H (max 1.2 g/day) +	Ciprofloxacin 10 - 15 mg/kg PO q12H (max 1500 mg/day) +	• Avoid Fluoroquinolones wherever possible in patients < 18 years old.

Infections and Likely Organisms	Suggested Treatment		Notes
	Recommended	**Alternative**	
	Clarithromycin 7.5 - 10 mg/kg (max 500 mg) PO q12H **OR** Azithromycin 10 - 12 mg/kg (max 500 mg/day) PO q24H	Amikacin 15 - 30 mg/kg/day IV/IM	
Talaromycosis (formerly Penicilliosis) *Talaromyces marneffei*			
Treatment	Amphotericin B Deoxycholate 0.6 - 1 mg/kg IV q24H for 14 days **Followed by** Itraconazole 3 - 5 mg/kg (max 200 mg) PO q24H		
Tuberculosis *Mycobacterium tuberculosis*			

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<tr><th>CHAPTER 4.11</th><th colspan="3">HUMAN IMMUNODEFICIENCY VIRUS-RELATED OPPORTUNISTIC INFECTIONS</th></tr>
<tr><th rowspan="2">Infections and Likely Organisms</th><th colspan="2">Suggested Treatment</th><th rowspan="2">Notes</th></tr>
<tr><th>Recommended</th><th>Alternative</th></tr>
<tr><td>Pulmonary disease</td><td>HRZ(E) for 2 months, followed by HR for 7 months</td><td></td><td rowspan="2">• 2nd line treatment / resistant mycobacterium TB – please consult ID specialist.
• Decision for duration of treatment depends on whether pulmonary or extrapulmonary TB.
• Intensive phase requires at least three drugs, and maintenance requires two drugs regimen.</td></tr>
<tr><td>Osseous or central nervous system disease</td><td>HRZE for 2 months, followed by HR for 10 months or longer</td><td></td></tr>
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Drug	Daily Dose Range (mg/kg)	Max. Daily Dose (mg)
Isoniazid, INH (H)	10 - 15	300
Rifampicin, RIF (R)	10 - 20	600
Pyrazinamide, PZA (Z)	20 - 40	2000
Ethambutol, EMB (E)	15 - 25	1000
Streptomycin, STR (S)	25 - 30	1000